

Lecture 2 & 3: Credit Curve and Fundamental Credit Analytics

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Yield Curves and Bond Analytics

- ▶ Yield curve is a term structure of interest rates (1M, 2M, 3M, 6M, 1Y, 2Y, 3Y, 5Y, 7Y, 10Y, 15Y, 30Y). (20Y?).
- ▶ From modeling perspective we can use zero rates, forward rates, discount factors or accumulation factors for a yield curve.
- ▶ Real world interest rate has its conventions in terms of interest calculation: rate convention, payment period length calculation, payment date etc.
- ▶ Duration and convexity defined using yield to maturity is good for single bond, but not good for portfolio or whole company asset liability management.
- ▶ DV01 defined as a sensitivity with respect to zero rates, forward rates are good for interest rate risk aggregation.
- ▶ For hedging purpose we tend to translate the sensitivity to the traded instruments input, this way we can quickly calculate the hedge ratio.

Credit Spread Curve

There are many different types of credit spreads: YTM credit spread, OAS, Z-spread, I-Spread, default swap spread. When a bond is priced to par, these measures are more or less very close to each other, but the deviation becomes larger when a bond is in deep discount (below 100 dollars) or with premium (above 100 dollars).

- **Credit spread or G-spread** (also called nominal spread) is the difference between yield on Treasury Bonds and yield on corporate bonds of same maturity. Because Treasury Bonds can be assumed to have zero default risk, the difference between yield on corporate bonds and Treasury bonds mainly represents the default risk, but also includes liquidity risk etc. We have another concept called *par yield*, it is yield when the bond is exactly priced to par (100 dollars).

$$\text{G-Spread} = Y_c - Y_g$$

Credit Spread

- **Z-spread** Z-spread stands for zero-volatility spread. It is the spread that must be added to each spot interest rate to cause the present value of the bond cash flows to equal the bond's price. Z-spread can be calculated by solving the following equation for Z :

$$P = \frac{CF_1}{(1 + Z_1 + S)} + \frac{CF_2}{(1 + Z_2 + S)^2} + \dots + \frac{CF_n}{(1 + Z_n + S)^n}$$

Where P is the price of the bond, CF_1 , CF_2 and CF_n are the first, second and n th cash flows, Z_1 , Z_2 and Z_n are the first, second and n th spot interest rate and S is the zero-volatility spread.

- **Asset Swap spread**: is spread over LIBOR if bond owner would like to exchange the bond's fixed cash flow into a floating cash flow, calculated by "LIBOR + asset swap spread".

Asset Swap Spread

Asset swap is an over-the-counter(OTC). It allows investment bankers to benchmark any fixed income securities with all features against the LIBOR market.

<http://corporatefinanceinstitute.com/resources/knowledge/trading-investing/asset-swap/> Asset swap spread is calculated as follows:

$$100 \sum_{i=1}^n (L_i + S) \Delta t_i d_{t_i} + \text{premium} = \sum_{i=1}^n C_i d_{t_i}$$

Credit Spread: OAS and I spread

- ▶ **Option-Adjusted Spread (OAS)** Option-adjusted spread equals zero-volatility spread minus the value of call option as stated in basis points. It is the appropriate yield measure for a callable bond: $OAS = Z - spread - OptionValue$.
- ▶ **I-spread** I-spread stands for interpolated spread. It is the difference between yield on a bond and the swap rate, i.e. the interest rate applicable to the fixed leg in the floating-for-fixed interest rate swap. The difference between yield on a bond and a benchmark curve such as LIBOR is useful in assessing credit risk of different bonds. Higher i-spread means higher credit risk. I-spread is typically lower than the G-spread.

Survival Analysis

Survival Time: from today to the time that default occurs .

We first introduce some fundamental concept in survival analysis to study default. We define a new random variable **time-until-default**, τ , which measures the time from today to the time that default occurs

$$F(t) = Pr[\tau \leq t], \quad S(t) = 1 - F(t) = Pr[\tau > t]$$

where $F(t)$ is the distribution function for τ , which gives the probability of default of an entity before time t , and $S(t)$ is called the **survival function**, which provides the probability of survival beyond t years. Another common entity used for survival analysis is called **hazard rate**. We calculate the following conditional default probability: Conditional on survival to the t years, the probability of default in the next Δt year:

$$Pr[t < \tau \leq t + \Delta t | \tau > t] = \frac{F(t + \Delta t) - F(t)}{1 - F(t)} = \frac{f(t)\Delta t}{1 - F(t)}$$

Hazard Rate

We define the hazard rate as follows:

$$h(t) = \frac{f(t)}{1 - F(t)} = -\frac{S'(t)}{S(t)}, \quad f(t) = F'(t).$$

which is the conditional instantaneous default probability conditional on survival to a future point of time. Once a hazard rate function is given, we can calculate any default or survival probability. For example,

$$S(t) = Pr[\tau > t] = e^{-\int_0^t h(s)ds}.$$

If $h(t) = h$ is a constant, then, $S(t) = e^{-ht}$, and $f(t) = he^{-ht}$, which is the density function for an exponential distribution. We very often use this assumption for the hazard rate during a small time interval. We sometimes use an annual conditional default or survival probabilities to express a credit curve such as $q_0, \dots, q_n$, or $p_0, \dots, p_n$ such as in the mortality table.

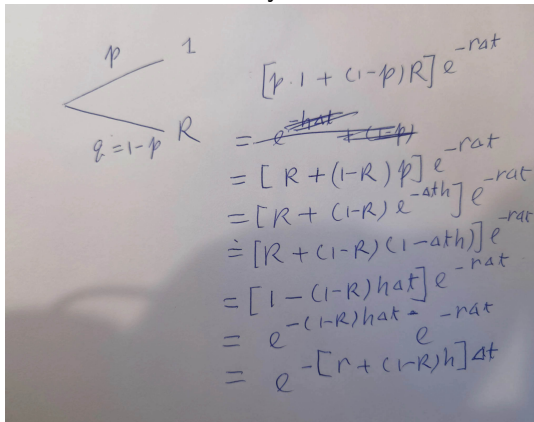
Credit Analytics

- ▶ how to describe default? We use a hazard rates or a *credit curve* to describe default
 1. Use a credit curve based on historical data for risk management
 2. Use a credit curve based on market information for pricing
- ▶ default treatment: what do you recover when a bond default?
 1. Legally, accrual interest plus face value
 2. Jarrow-Turnbull, future cash flows discounted by risk free interest rate
 3. Duffie-Singleton, market price just before default(default is an unpredictable event)
- ▶ recovery rate: Recovery rate is quite volatile, in practice we usually use an assumption of mean recovery rate or a random recovery rate such as beta distribution

One Period Credit Valuation

Suppose we have a zero coupon bond which matures at the end of the period $[0, \Delta t]$.

If the bond does not default, you receive \$1 dollar, if it defaults, you receive a recovery



The image shows a handwritten derivation of the credit-adjusted bond price. On the left, a binomial tree node is shown with an upward branch labeled p leading to a value of 1, and a downward branch labeled $q = 1 - p$ leading to a value of R . To the right of the tree, the following equations are written:

$$\begin{aligned}
 & [p \cdot 1 + (1-p)R] e^{-rat} \\
 &= \cancel{e^{-\cancel{hat} + (1-p)}} e^{-rat} \\
 &= [R + (1-R)p] e^{-rat} \\
 &= [R + (1-R)e^{-ath}] e^{-rat} \\
 &\doteq [R + (1-R)(1-ath)] e^{-rat} \\
 &= [1 - (1-R)hat] e^{-rat} \\
 &= e^{-(1-R)hat - rat} \\
 &= e^{-[r + (1-R)h] \Delta t}
 \end{aligned}$$

rate R

This shows that a defaultable cash flow can be valued as if it were default free, but adjusting the short rate by $(1 - R) \times h$, which can be interpreted as *expected loss*.

Multi-period Credit Valuation Framework

Suppose we have a risky cash flow stream $C_1, C_2, \dots, C_n$ at times $t_1, t_2, \dots, t_n$. Considering the period $[t_i, t_{i+1}]$, using Duffie-Singleton assumption, if default, we would have $R_{i+1}[C_{i+1} + V(t_{i+1})]$, then we have the following recursive formula:

$$\begin{aligned} V(t_i) &= D(t_i, t_{i+1})[p_i(C_{i+1} + V(t_{i+1}))] + (1 - p_i)R(t_{i+1})(C_{i+1} + V(t_{i+1}))] \\ &= [p_i + (1 - p_i)R(t_{i+1})][C_{i+1} + V(t_{i+1})] \end{aligned}$$

where $i = 0, 1, \dots, n$ and the boundary condition is $V_{t_n} = 0$, and $D(t_i, t_{i+1})$ is the forward discount factor which is the value at time t_i for \$1 payment at time t_{i+1} . By consecutive substitution, we have

$$V(t_0) = \sum_{i=1}^n D(t_0, t_i) \left(\prod_{j=0}^{i-1} [p_j + (1 - p_j)R(t_{j+1})] \right) C_i$$

Multi-period Credit Valuation Framework

If we denote

$$D^c(t_0, t_i) = \prod_{j=0}^{i-1} [p_j + (1 - p_j)R(t_{j+1})] = e^{-\int_0^{t_i} (1-R(s))h(s)ds}$$

as the **credit discount factor**, then we have

$$V(t_0) = \sum_{i=1}^n D(t_0, t_i) D^c(t_0, t_i) C_i = \sum_{i=1}^n e^{-\int_0^{t_i} [r(s) + (1-R(s))h(s)]ds} C_i$$

This shows that we can value risky cash flows as they were risk free, by multiplying a credit count factor. This credit discount factor depends on both default probabilities and recovery. Remember, this is based on

1. deterministic interest rate, and credit curve, and recovery.
2. Duffie-Singleton recovery treatment assumption

Multi-period Credit Valuation Framework

The question is: does this formulation work for more general case? The answer is Yes!
In a more general case when interest rate, and hazard rate are **stochastic**, the general valuation formula for a risky cash flow of \$1 is

$$V = E \left[e^{-\int_0^t [r(s) + h(s)(1-R(s))] ds} \right]$$

This is the main result from Duffie-Singleton paper.

References

- ▶ David X. Li, Constructing a Credit Curve
- ▶ Duffie, D and Singleton, K., Modeling Term Structures of Defaultable Bonds1
<https://web.stanford.edu/~duffie/ds.pdf>